

Duy Nguyen

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Summary

Data scientist and MS candidate who designs and runs rigorous ML experiments—and builds the data pipelines they run on. Recent research isolated the retrieval *encoder*, not decoder fine-tuning, as the lever in medical-imaging RAG; the result is a first-author paper accepted at PSB 2027 with an oral presentation. I own problems end to end: experiment design, modeling, evaluation, and reporting. The part I am most careful about is validity: my replication caught a mislabeled metric in the published benchmark, and my own headline result is anchored by a negative control.

Education

M.S. in Data Science — Seattle University

Expected Jun 2027

Seattle, WA

Certificate, Machine Learning & AI — UC Berkeley (Online)

Jul 2024

B.A. in Economics, Data Analysis Concentration — Simon Fraser University

May 2023

Burnaby, BC, Canada

Experience

Graduate Research Assistant — Medical Vision–Language RAG

Spring 2026 – Present

Seattle University / Advisor: Dr. Wenjing Yang

- Replicated and audited a published mammography-RAG benchmark on VinDr-Mammo (20,000 images / 5,000 patients) in **Python** and **PyTorch**, and found its headline scores mislabeled — weighted metrics reported as “macro” — correcting the basis on which every subsequent comparison rests.
- Designed and ran a 24-cell controlled experiment (4 encoder substrates $\times$ frozen/fine-tuned $\times$ 3 vision–language models) with **HuggingFace Transformers**, **ChromaDB**, and **scikit-learn** to isolate the causal driver of generated-report quality.
- Isolated the retrieval *encoder* as the lever: replacing the frozen OpenCLIP retriever with a domain-native, supervised-contrastively fine-tuned Mammo-CLIP encoder—decoder frozen throughout—raised suspicion macro-F1 0.35 to 0.48 (+**0.13**) and BI-RADS (+0.10) on LLaVA-Med; the fine-tuning step alone accounts for +0.11 (Qwen2.5-VL) and +0.06 (LLaVA-Med).
- Designed the study to be able to falsify its own conclusion: encoder substrate and frozen/fine-tuned were varied as independent factors so the causal attribution could not rest on a single comparison, and retrieval-insensitive MedGemma was carried throughout as a **negative control**.
- Automated evaluation with reproducible, self-verifying generators (**Python/pandas**) that recompute 100+ reported metrics byte-exact from raw model predictions; authored a 37-page report, condensed into a first-author paper accepted at the Pacific Symposium on Biocomputing (PSB) 2027, with an oral presentation (Jan 2027).

Research Data Engineer

Spring 2026 – Present

Computational Neuroscience Research Group (Prof. Brian Fischer), Seattle University

- Supported an NIH CRCNS-funded computational neuroscience project on barn owl sound localization by building data infrastructure for neural recordings (full-time over the summer, continuing).
- Engineered an end-to-end **ETL** pipeline in **Python** (**pandas**, **SciPy**, **h5py**) converting 5+ proprietary electrophysiology formats (XDPHYS .iid/.itd/.bf, MATLAB .mat, headerless CSV) into a normalized **SQLite** database of 195 neurons and 1,325 recording passes, with a second researcher’s spike-reliability dataset staged and schema-provisioned for load.
- Designed the relational schema and ER model (documented in **LaTeX**) with foreign-key constraints and explicit data-quality rules (missing-data handling, tracked exclusions, path remapping), consolidating scattered lab files into one queryable source of truth.
- Tuned queries and ETL to meet <200 ms query and <30 s full-load targets, letting lab students self-serve neuron-level queries that previously required manual file handling.
- Ran **SQL/SQLite** schema and data-integrity audits that caught and corrected issues (all-zero rows, weak-tuning exclusions) before downstream analysis.

- “Will AI Make Human Work Worthless—or Priceless?”: a nine-act interactive data story (**D3.js**, **Scrollama.js**) with scroll-driven animation and a reader-controlled parameter widget, translating an original nested-CES general-equilibrium model of AI and cognitive labor into a narrative for non-specialists. Judged on narrative, data interpretation, scrollytelling, and visual design.

Skills

Programming: Python, SQL, JavaScript

ML / AI: PyTorch, HuggingFace Transformers, scikit-learn, Retrieval-Augmented Generation (RAG), vision-language models, QLoRA / parameter-efficient fine-tuning

Data: pandas, NumPy, SciPy, SQLite, ETL design, relational schema / ER design, data quality

Experimentation & Causal Inference: experimental design, factorial designs, controls and negative controls, A/B testing, hypothesis testing, effect-size estimation, replication and reproducibility auditing

Tools: Jupyter, Git, LaTeX, D3.js, Scrollama.js, Google Colab (A100 GPU), Ollama